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[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

May-26-0206

MCA-6204 (Computer Networks)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all by selecting one question each from section A, B, C and D. Section-E is compulsory.

SECTION - A

1. (a) Explain the OSI Reference Model with a neat diagram. (8)
- (b) A digital communication system has a bandwidth of 1 MHz and an SNR of 40 dB. Calculate the maximum data rate using Shannon's Capacity Theorem. (4)
2. (a) Discuss different types of network topologies and their advantages. (8)
- (b) Compare and contrast OSI and TCP/IP model. (4)

SECTION - B

3. (a) Describe Stop-and-Wait ARQ and Go-Back-N ARQ protocols with their working principles. (6)
- (b) What is HDLC Protocol? Explain its frame structure and working. (6)
4. (a) Explain the working of CSMA/CA protocol. How do it handle collisions? (6)

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- (b) Differentiate between IEEE 802.3, IEEE 802.5, and IEEE 802.11 in terms of standards, transmission methods, and applications. (6)

SECTION - C

5. (a) Describe distance vector routing algorithm. How does Dijkstra's algorithm help in shortest path routing? (6)
- (b) What is congestion control in networking? Explain different congestion control techniques. (6)
6. (a) What is Quality of Service (QoS) in computer networks? (6)
- (b) Explain flooding and broadcasting in routing. How do they differ? (6)

SECTION - D

7. (a) What is BGP? Explain its significance in inter-domain routing. (6)
- (b) A company is assigned the IP address 192.168.10.0/24 and needs to divide it into four subnets for different departments. What will be the new subnet mask? Determine the IP range and broadcast address for each subnet. Calculate the maximum number of usable hosts per subnet. (6)
8. (a) Discuss the functions of the Application Layer Protocols: DNS, SMTP, FTP, and TELNET. (8)
- (b) What are the different performance metrics used in network analysis? (4)

[P.T.O.]

SECTION - E (Compulsory)
(Short Answer Questions)

9. Explain the following:

- (a) Define bandwidth and latency in computer networks.
- (b) Define ICMP and its role in networking.
- (c) Differentiate between circuit switching and packet switching.
- (d) Compare IPv4 and IPv6 addressing schemes.
- (e) Explain the concept of fragmentation in networkings.
- (f) What is Tunneling in computer networks? (6×2=12)